Intelligent Systems with Applications 20 (2023) 200265

Contents lists available at ScienceDirect

Intelligent Systems with Applications

journal homepage: www.journals.elsevier.com/intelligent-systems-with-applications

Analysis of the application of path finding system based on efficiency improvement in smart tourism

Shuping Liu [a], Hao Wu [b,*]

[a] College of Rural Revitalization, Zhengzhou Vocational University of Information and Technology, Zhengzhou, 450008, China
[b] Party Committee, Zhengzhou Vocational University of Information and Technology, Zhengzhou, 450008, China

ARTICLE INFO

Keywords:
Convergence efficiency
Ant colony algorithm
Genetic algorithm
Transfer probability
Tourist route planning

ABSTRACT

With the proposal and development of intelligent tourism destination, the problem of tourism route planning has become a hot topic in tourism research. This study proposes an intelligent tourism route planning model based on efficiency improvement for Guangxi tourism route planning. In order to improve the efficiency of the model, this paper combines ant colony algorithm with genetic algorithm and introduces smoothing factor and guiding factor, and obtains genetic algorithm (genetic algorithm-path smoothing ant colony optimization) with changing probability. Experimental results show that the convergence speed of the algorithm is fast, and it can converge in 78 iterations, and the operation efficiency is higher. The result error of the proposed algorithm is 1.39, which is smaller than that of other comparative algorithms. In the actual application effect evaluation, the average satisfaction of the planned route of the intelligent tourism route planning model based on efficiency improvement is 15.38, which is higher than that of the traditional genetic algorithm and simulated annealing algorithm. It shows that the model can not only efficiently plan the best travel route, but also comprehensively consider the whole trip of tourists, so that tourists can have a better travel experience. The research provides an intelligent route planning method that is more efficient and more in line with tourists' needs for tourism development.

Introduction

With the growth of the Chinese economy, the development of the Internet, and the improvement of travel conditions, it has led to an increasing preference for free-wheeling forms of travel, such as self-driving tours, RV tours, and backpacking (Zhang et al., 2020). Especially after the epidemic, the economy has recovered and the tourism industry has taken on a new vigor. Currently, there is even a rise in the special forces-style and military training style of traveling to several sites in one day, putting forward higher requirements for personalized travel itinerary planning (Song et al., 2022). When tourists plan their travel itineraries independently, they need to obtain relevant contents from a large amount of network information. The problem of tourist information overload caused by the network sharing mechanism and the various constraints that need to be considered in the itinerary make tourists spend a lot of energy in screening the appropriate tourist services. This is certainly a complex dilemma for tourists visiting unfamiliar cities. If an optimal travel route can be quickly planned for a journey with many attractions and a short duration, it will not only enhance the consumer's journey experience, but also be an unbeatable benefit for the attractions.

The development of computer science and technology in recent years has provided new ideas and methods to solve the travel route planning problem (Zan, 2022). Many scholars have studied the travel route problem through existing algorithms, including ant colony algorithm, genetic algorithm, particle swarm algorithm and simulated annealing algorithm (Mohammadzadeh & Gharehchopogh, 2020; Gharehchopogh et al., 2023a; Gharehchopogh, 2023). The Ant colony algorithm is a probabilistic algorithm used to find the optimal path. It was inspired by the way ants find their way around food. This algorithm has the characteristics of distributed computation, positive feedback of information and heuristic search. It is a heuristic global optimization algorithm in evolutionary algorithms (Gharehchopogh et al., 2023b).

This research fully absorbs the previous work achievements, and proposes an intelligent tourism route planning model that combines Ant colony optimization algorithms and genetic algorithm, and improves the global search ability through cross operation and roulette wheel selection. At the same time, smoothing and guiding factors are introduced to improve the smoothness and operational efficiency of the model. The main contribution of this research is to propose a new combination algorithm for path optimization, which provides ideas and methods for

* Corresponding author.
E-mail address: wuhao014@hotmail.com (H. Wu).

https://doi.org/10.1016/j.iswa.2023.200265
Received 22 May 2023; Received in revised form 19 July 2023; Accepted 8 August 2023
Available online 9 August 2023
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research on improving the performance of Algorithmic efficiency, global search capability, etc. At the same time, it also provides a feasible path planning model for freelance tourists, which can quickly plan the optimal travel path and time in complex environments, reduce tourists' energy consumption, and improve their travel quality and experience.

The first part of the research describes the research status of path planning and Ant colony optimization algorithms, and proposes improvement methods according to the shortcomings of Ant colony optimization algorithms in path planning. The second part describes in detail the path optimization process of the improved Ant colony optimization algorithms and the construction process of the multi-objective tourism route planning model based on the algorithm. The third part evaluates the efficiency and performance of the path planning of the improved Ant colony optimization algorithms, and compares the application effect of the tourism route planning model. The fourth part summarizes the research process and experimental results, and proposes improvement directions to address the shortcomings.

Related works

Path planning belongs to optimization problem, there are many academic research on optimization algorithm. An improved PSOBSA algorithm is proposed by Zaman et al. The algorithm combines the backtracking search optimization algorithm (BSA) and particle swarm optimization search (PSO), uses BSA to improve the global search capability of PSO, and introduces a new mutation operator to improve the convergence accuracy. Experimental results showed that the proposed algorithm outperformed other well-known heuristic and several recent PSO variants in global search capability, accuracy and convergence speed (Zaman and Gharehchopogh, 2021). Shishavan et al. improved the cuckoo search optimization (CSO) algorithm by using genetic algorithm (GA) to solve the community discovery problem in complex networks. The algorithm uses GA operators to improve the speed and accuracy of CSO. Experimental results showed that the proposed algorithm had advantages in modularization and NMI. Compared to other algorithms, the proposed algorithm achieves an average improvement of 54% in modularity and 88% in NMI. The problems of premature convergence, delayed convergence and local optimization of the CSO algorithm are solved, which is beneficial for finding communities in complex networks (Shishavan & Gharehchopogh, 2022). The human whale optimization algorithm (WOA) of Shen et al. tends to converge slowly and easily falls into local optimization. In this paper, a mutation whale optimization algorithm (MEWOA) based on multi-population evolution is proposed to replace WOA. First, the individual is divided into three equal-sized subpopulations according to the individual's fitness. Second, different mechanisms are used to assign the movement strategy of each sub-population. Finally, a new population evolution strategy is introduced to help MEWOA improve its global optimization ability and avoid falling into local optimization. The experimental results showed that MEWOA had faster convergence speed, shorter running time and higher accuracy than other algorithms on most reference functions, and can solve the design problem of capability engineering. This study provides a new avenue for WOA optimization (Shen et al., 2023).

In the context of combinatorial optimization problems such as path planning, ant colony algorithms have been favored by many researchers for their high speed, high accuracy and high efficiency. Mutar et al. proposed an improved ant colony system for solving the capacity vehicle path problem. The method sets the condition function according to the demand of capacity vehicle delivery customers, and seeks the best capacity vehicle path with the shortest path as the evaluation index. Comparative experiments showed that the method gave better results (Mutar et al., 2020). Yasear et al. used Harris' hawk optimizer to optimize the pheromone coefficients and evaporation rate, heuristic coefficients, decision rules and other parameters of the ant colony algorithm system to better solve the traveler problem. The improved algorithm can find the optimal solution in bayg29, att48, and berlin52 data and shows excellent performance in solving the traveler problem compared with other algorithms (Yasear & Ku-Mahamud, 2021). Li et al. proposed to introduce the "distance" small habitat technique and cyclic Jaccard distance metric. The experimental results showed that the algorithm had better performance in terms of quality and diversity of optimal solutions (Li et al., 2022). Xie et al. proposed a path planning model for manned submersible based on the ant colony algorithm, which calculates the probability density of obstacles through a quantitative method of artificial potential field rejection, and uses the multi-angle path quality evaluation function as the iteration and update command of the model to find the optimal path. Experiments have shown that the path planned by this algorithm has better results in reducing driving difficulty and energy consumption and improving navigation safety (Xie & Shi, 2021). The remote delivery application of UAVs was the focus of Shao et al.'s proposal for a UAV path planning system integrating the A* algorithm and ant colony algorithm. The authors set special penalty values as the objective function to obtain both the shortest path length and the fewest possible landing stations in order to efficiently solve the optimal path. The algorithm was experimented with 10 random distribution cases and the comparison results showed the superiority of the algorithm (Shao et al., 2020). Ambewadkar et al. optimized the CMM flatness error detection by solving the shortest probe path for production inspection by an improved ant colony algorithm, thus reducing the time for flatness inspection. Experimental results showed that the method was effective and performs optimally in comparative experiments (Ambewadkar & Gajre, 2021). Wu et al. proposed to combine the ant colony algorithm with a manual correction algorithm that adds an angle calculation operator to improve the global search capability and navigation of the path in the path navigation study of UAVs. Simulation experiments showed that the path obtained by this method was smooth and close to the optimal path (Wu & Wei, 2021). In order to improve the search efficiency and convergence speed of the ant colony algorithm, Zheng et al. introduced an adaptive heuristic function in the algorithm, which automatically uses the shortest distance passed by ants to update the heuristic information. Meanwhile, the update strategy of local pheromone is optimized by combining the reward and punishment rules, and the pseudo-random state transfer rule is introduced to optimize the state transfer function. Simulation comparison experiments showed that the algorithm had better results in path planning (Zheng et al., 2020).

In summary, the experimental results of the ant colony algorithm in path planning are all good. Therefore, the study plans to optimize the traditional ant colony algorithm, integrate the ant colony algorithm with genetic algorithm, and introduce the smoothing factor and bootstrap factor to improve the operation efficiency of the model. Finally, the effectiveness of the algorithm is evaluated by path length and convergence speed, and it is used for the first time in the planning of tourist routes in Guangxi, and the practicality and superiority of the algorithm are verified by comparison experiments.

Construction of intelligent planning model for tourism routes

This part constructs the shortest collision free path planning model based on Ant colony optimization algorithms, and then optimizes the problems of Ant colony optimization algorithms such as insufficient global search ability and low operation efficiency by combining genetic algorithm, smoothing factor and guiding factor. The tourism route planning model is constructed by comprehensively considering various influencing factors, and the best path is solved by improving the Ant colony optimization algorithms.

Improved ant colony algorithm for collision-free path planning

The planning of a personalized travel itinerary requires consideration of several factors such as tourist preferences, transportation

conditions and travel time. This process can be regarded as a combinatorial optimization problem with multiple constraints. Ant colony optimization (ACO) is usually chosen for solving such problems. ACO reduces the travel route planning problem to a path finding process, i.e., finding the shortest collision-free path in a map with obstacles (Yang, 2022). The path optimization process of the ant colony algorithm is shown in Fig. 1. First, grid maps are constructed, then the heuristic information matrix is constructed, and the path, path length, heuristic information matrix and tabu list are updated. The colony compares the path with the original shortest path, and if the path is longer than the shortest path, it continues to find the path. Instead, the ant colony finds the shortest path and the algorithm stops iterating.

First, all parameters of the model are initialized. Then the obstacle raster map with y rows and x columns is created, and the rasters in the map are marked with the serial number $(1, 2, \cdots, i)$ from left to right and from bottom to top. If the side length of each grid is a, the coordinate position of the i grid can be calculated by Eq. (1).

$$\begin{cases} x_i = a * [\bmod (i, y) - a/2] \\ y_i = a * [x + a/2 - \mathrm{ceil}(i, x)] \end{cases} \tag{1}$$

The mod and ceil represent the remainder function and the upward rounding function, respectively. For the movement of ants in the raster map, an eight-way steering strategy is used, i.e., ants can move from the i raster to the adjacent eight rasters: top, bottom, left, right, top-right, bottom-right, top-left, and bottom-left. Then based on the grid coordinates obtained from Eq. (1), the D_{ij} which is the distance between the i-th grid and the adjacent grid j, can be calculated as shown in Eq. (2).

$$D_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{2}$$

To make the ants move in a path avoiding obstacles, a heuristic information matrix D is introduced to evaluate the feasibility of traveling from each grid to the adjacent grid. The heuristic factor is assigned to $1/D_{ij}$ for the unobstructed grid of i and 0 for the obstructed grid of j, and the matrix D is entered. Then the ants can determine whether the eight neighboring grids in D are obstacles or not based on the data in D. Since the trajectory of ants in the ant colony algorithm is randomly distributed, the probability of going from grid i to grid j in t iterations is influenced by the pheromone $a_{ij}(t)$ and the heuristic information. The relevant formula is given in Eq. (3).

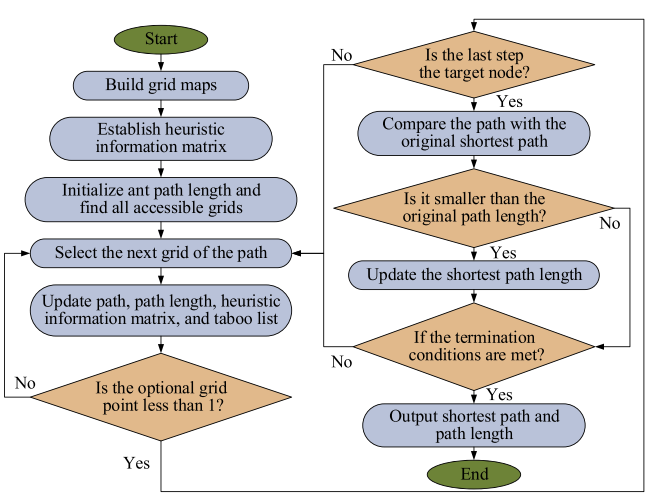


Fig. 1. Flowchart of path finding optimization of ant colony algorithm.

$$p_{ij}^k(t) = \begin{cases} \dfrac{[a_{ij}(t)]^\alpha * [1/D_{ij}(t)]^\beta}{\sum\limits_{j \in J_k(i)} [a_{ij}(t)]^\alpha * [1/D_{ij}(t)]^\beta}, j \in J_k(i) \\ 0, \text{Other} \end{cases} \tag{3}$$

In Eq. (3), $p_{ij}^k(t)$ is the probability that the k-th ant moves from grid i to grid j in t iterations; α and β are the pheromone weights and heuristic information weights; $J_k(i)$ is the set of the next movement directions that ants k can choose at grid i. The initial value of the pheromone can be set artificially and will be updated and changed with the number of iterations. Therefore, the formula for calculating the pheromone $a_{ij}(t+n)$ between the raster i and j marked by the ant k during the iterations from t to $t+n$ is shown in Eq. (4).

$$a_{ij}(t+n) = (1-\rho) * a_{ij}(t) + \sum_1^k \Delta a_{ij}^k \tag{4}$$

In Eq. (4), ρ takes values between [0,1], which indicates the degree of pheromone dissipation, and Δa_{ij}^k indicates the amount of pheromone change of the ant k, which can be described by segmentation in Eq. (5). Where Q and L_k denote the controllable parameters and the length of the path traveled by the ant k.

$$\Delta a_{ij}^k = \begin{cases} Q/L_k, & \text{Passed through } ij \\ 0, & \text{Didn't pass through } ij \end{cases} \tag{5}$$

From equations (3) to (5), it can be seen that if more pheromones are available for a particular route, the higher the heuristic factor, the greater the probability that the ants will choose that route and eventually reach optimal path planning. In the iterative process, to prevent the ants from repeatedly visiting the complete path already taken, a forbidden table is created for each ant and its value is initialized to 1. The path length of the ants is also initialized, and iteration and updating are performed according to the process shown in Fig. 1. Since the algorithm requires more iterations to run and is less efficient, and the probability calculation has a greater impact on the number of iterations, the probability calculation part is optimized (Plithogeny, 2022). By introducing the smoothing factor $\omega_{ij}(t)$ and the bootstrap factor $\lambda_{ij}(t)$, the probability calculation formula becomes Eq. (6). Where φ and γ are the weights of $\lambda_{ij}(t)$ and $\omega_{ij}(t)$.

$$p_{ij}^k(t) = \begin{cases} \dfrac{[a_{ij}(t)]^\alpha * [1/D_{ij}(t)]^\beta [\lambda_{ij}(t)]^\varphi [\omega_{ij}(t)]^\gamma}{\sum\limits_{j \in J_k(i)} [a_{ij}(t)]^\alpha * [1/D_{ij}(t)]^\beta [\lambda_{ij}(t)]^\varphi [\omega_{ij}(t)]^\gamma}, j \in J_k(i) \\ 0, \text{Other} \end{cases} \tag{6}$$

Increasing the number of turning angles and reducing the number of turns can effectively shorten the distance of the path (Wang et al., 2022). The introduction of smoothing factor to smooth the corner θ not only makes the path appear smoother, but also allows the algorithm to find the optimal path quickly in fewer iterations. Its calculation formula is given in Eq. (7).

$$\omega_{ij}(t) = \begin{cases} 1, \theta = 0^\circ \\ 2, \theta = 45^\circ \\ 3, \theta = 90^\circ \\ 4, \theta = 135^\circ \end{cases} \tag{7}$$

The role of pheromone and heuristic factor increases with the number of iterations, and the role of both is very small at the beginning of the algorithm operation. This is one of the factors that lead to the increase of the number of iterations. The introduction of the bootstrap factor can help the ants clarify the general direction of the path in the early stage of the operation and improve the operation efficiency. Its calculation formula is shown in Eq. (8).

$$\lambda_{ij}(t) = \frac{k}{K} \cdot \frac{t}{t_{\max}} \cdot \frac{1}{D_{i,\mathrm{E}} + D_{ij}} \tag{8}$$

In Eq. (8), t and $t_{\max}$ are the current iteration and the maximum iteration; K and k are the total number of ants and the number of ants under t iterations; $D_{i,E}$ and $D_{i,j}$ represent the distance from the grid i to the end point e and the grid j. From Eq. (8), it can be seen that the nearly exponential growth of the bootstrap factor allows the ants to complete the path search process faster. In addition, to improve the overall operational efficiency of the model, the study also incorporates genetic algorithm (GA) based on the ant colony algorithm (Wang & Xue, 2022). The initial pheromone was obtained by individuals of the genetic algorithm during the initialization of the pheromone. Then the routes taken by the ants are recorded using cellular structures instead of taboo tables. By crossover operation of the genetic algorithm, some of the routes taken by the ants are crossed over to obtain new individuals. This method can expand the population sample with less number of iterations. Finally, the roulette wheel selection method of the genetic algorithm is introduced to find the probability of an individual being selected according to the ratio of individual fitness to the sum of fitness of all individuals in the population. And the individuals with probability greater than the set value are used as new individuals to form a new population. The process of ant colony algorithm (Genetic algorithm-Path smoothing ant colony optimization, GA-PSACO) that changes the probability and incorporates genetic algorithm is shown in Fig. 2. The algorithm first builds a grid map, finds all accessible grids for the next step, and then records each ant's path with a cellular structure. The algorithm combines selection and crossover operations with genetic algorithms to calculate the path length, update the ant state and the pheromone matrix until the target node is reached. The path is then compared with the original shortest path, and if it is shorter than the original path length, the shortest path length is updated. Otherwise, iteration continues until the termination condition is met and the shortest path and path length are output.

Tourism route planning model considering multiple factors

Tourism route planning mainly focuses on the situation that tourists travel two or more attractions in a period of time, and factors such as distance and time have a large impact on tourists' choices. Therefore, tourism route planning can be regarded as a multi-objective path finding process based on ant colony algorithm (Chen et al., 2021). In recent years, scholars have carried out many studies around tourism route planning, but most of them consider fewer influencing factors, which makes the model much less valuable in practical applications. This study, on the other hand, integrated factors such as travel time, travel distance, crowding, preference degree, and dwell time (Liu et al., 2016), and designed a multi-objective tourism route planning model based on an improved ant colony algorithm. With tourist satisfaction as the evaluation index, an optimal tourist route is planned under a variety of condition constraints. First, the distribution map of tourist attractions is constructed on the raster map of the improved ant colony algorithm, as shown in Fig. 3.

From Fig. 3, it can be seen that the shortest path output according to the algorithm iteration is the optimal travel path by defining i and j in the algorithm as the current attraction and the next one that can be reached without considering other factors. In contrast, many constraints exist in reality, and the obtained travel routes differ due to the different distribution of condition weights. The constraints such as trip time factor, congestion factor and preference factor are explained in turn. Travel time is the time consumed on the journey from the current attraction i to the next attraction j, and is related to the ease of transportation between the two places as well as personal state and rest time. The latter two are highly influenced by the tourists themselves and cannot be used as a qualitative reference, so only the travel time is calculated based on the ease of transportation between attractions. Assuming $M_{i,j}$ is the travel time factor, the relevant calculation formula is given in Eq. (9).

$$M_{i,j} = \frac{D_{i,j}}{V_{i,j}} \tag{9}$$

Where $D_{i,j}$ and $V_{i,j}$ are the distance and speed of visitors from the current attraction i to the next attraction j, respectively. By scaling down the attraction distribution map to a raster map, $D_{i,j}$ can be calculated from Eq. (2). $V_{i,j}$ can be assigned according to the existing urban traffic conditions. If i and j are both convenient, then the speed is the fastest and $V_{i,j}$ is the largest. If only i or j is convenient, then the speed is the second fastest and $V_{i,j}$ is the middle. If i and j are both inconvenient, then the speed is the slowest and $V_{i,j}$ is the smallest. Crowding is determined by the maximum visitor capacity of the attraction and the number of visitors (Majumdar et al., 2021), which is calculated by the formula shown in Eq. (10).

$$C_j = \frac{E_j}{A_j} \tag{10}$$

In Eq. (10), C_j denotes the visitor crowding factor of the attraction j. E_j and A_j are the number of tourists traveling at the attraction j and the maximum tourist carrying capacity of the attraction. E_j can be obtained

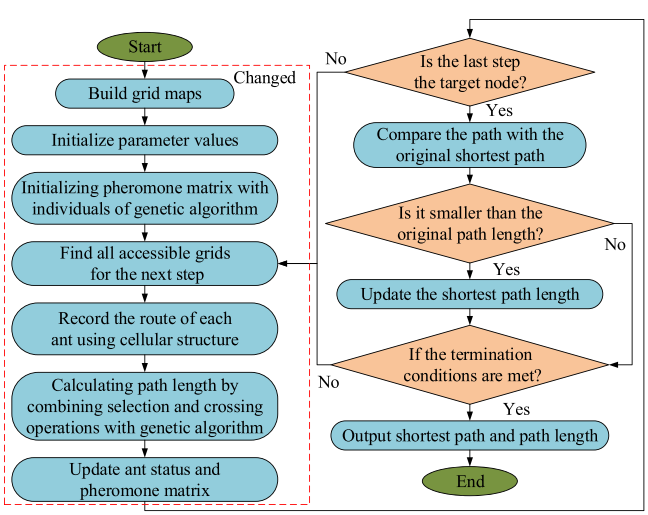


Fig. 2. Flowchart of GA-PSACO algorithm.

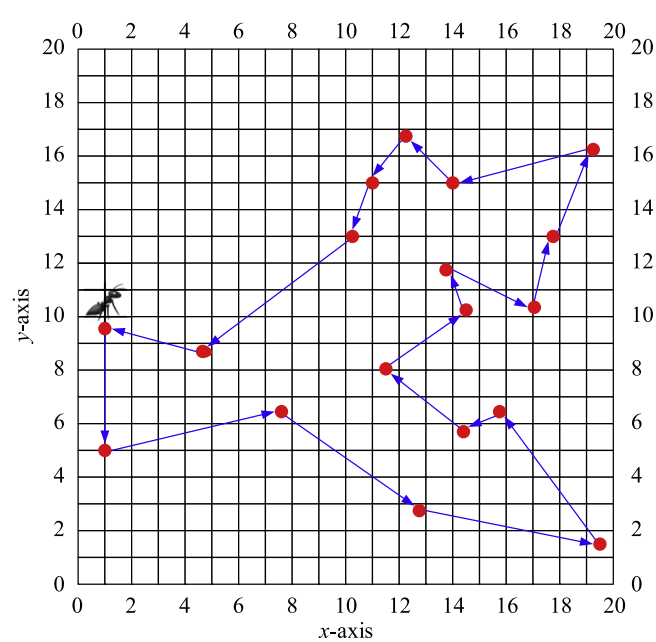


Fig. 3. Simulation of travel route considering only the shortest path.

from the historical tourism data of the attraction j and A_j is set based on the environmental carrying capacity given by the attraction j . The degree of tourists' preference for the attraction j can be described by the preference factor H_j, which depends mainly on the personalized recommendations of the network and tourists' own preferences (Ahmad et al., 2019), and the relevant formula is given in Eq. (11).

$$H_j = s_j * g_j \tag{11}$$

In Eq. (11) s_j indicates the similarity of the visitor's preferred characteristics to the attraction j, determined by the similarity of the desired label entered by the visitor to the label of the attraction j. g_j is the average rating of the attraction j on the web. According to the calculation of Eq. (6), the probability of a visitor traveling from attraction i to attraction j can be found by combining the three influence factors mentioned above, and the result $P_{i,j}$ is shown in Eq. (12).

$$P_{i,j} = \frac{\left[1/M_{i,j}\right]^{\varepsilon} * \left[1/C_j\right]^{\tau} * H_j^{\delta}}{\sum\limits_{j \in J_k(i)} \left[1/M_{i,j}\right]^{\varepsilon} * \left[1/C_j\right]^{\tau} * H_j^{\delta}} \tag{12}$$

In Eq. (12) ε, τ and δ are the influence weights of trip time factor, crowdedness, and preference level, respectively. Also based on the crowding level of the attraction j, the distance traveled and the visitors' preference level of the attraction j, the visitors' satisfaction level r_j of the attraction j can be calculated from Eq. (13) .

$$r_j = H_j / \left(M_{i,j} * C_j\right) \tag{13}$$

Taking the maximum satisfaction of tourists with the tour route R as the evaluation index of path finding (Canalejo & Rio, 2018), the result as shown in Eq. (14) can be obtained.

$$R = \max\left(\sum_{i=1}^{n-1} \sum_{j=1}^{n-1} z_i r_i\right) \tag{14}$$

In Eq. (14), z_i takes the value of 0 or 1, where 0 means that the visitor has not visited the site i and 1 means that the site i has been visited. To avoid repeated visits to sites and routes already visited, R needs to satisfy the constraints shown in Eq. (15).

$$\begin{cases} \sum\limits_{i=1}^{n-1} z_{i,l} = \sum\limits_{j=1}^{n-1} z_{l,j} \le 1, (i \ne l, j \ne l) \\ \sum\limits_{j \in J_k(i)} z_{i,j} \le |J_k(i)| - 1, (\forall J_k(i) \subset U, 3 \le |J_k(i)| \le n) \end{cases} \tag{15}$$

Where $|J_k(i)|$ indicates the number of attractions in the set $J_k(i)$; $z_{i,j}$, $z_{i,l}$, $z_{l,j}$ takes the value of 0 means that the visitor has not passed through the paths i-j, i-l and L-j, and takes the value of 1 means that the paths i-j, i-l and L-j have been passed; U indicates the set of all attractions. The planning of paths also needs to give the tour time to facilitate visitors to allocate their trips according to time. The tour time is mainly related to the tourists' preference, and its related formula is shown in Eq. (16).

$$T_j = T \cdot \frac{H_j^{3/2}}{\sum H_j^{3/2}} \tag{16}$$

In Eq. (16) T_j and T are the allocated tour time and the total tour time at the attraction j, where T can be set according to the actual demand of tourists. Combining all the above cases again for the route planning of the attractions in Fig. 3, the optimal tour route obtained is shown in Fig. 4.

Performance evaluation of intelligent planning model for tourism routes

This section tests the effectiveness of the proposed algorithm in path planning in simple grid maps and verifies its operational advantages on the Eil51 dataset. Then, taking Guangxi as an example, the real value is

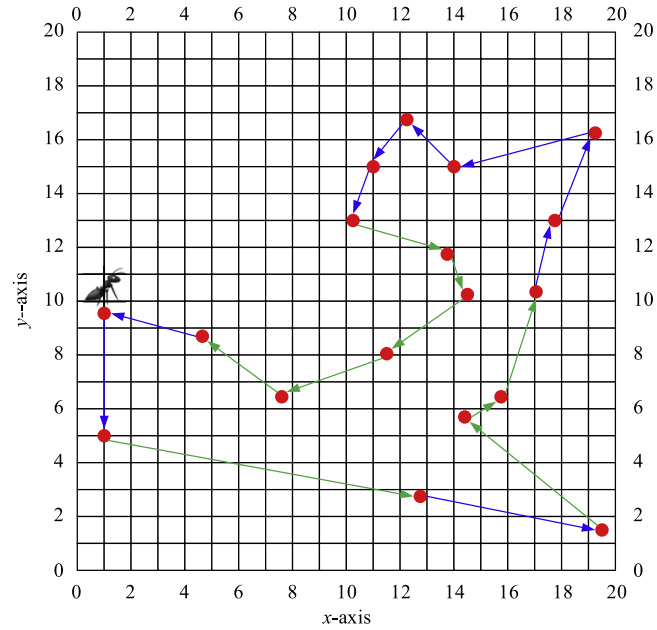


Fig. 4. Simulation of the tourist route obtained by combining all the scenarios.

used to simulate the tourism path planning model, which is solved by the improved Ant colony optimization algorithms and several other different algorithms. Through comparative analysis, the advantages of the improved Ant colony optimization algorithms in tourism path planning are analyzed.

Efficiency evaluation of path planning model with improved ant colony algorithm

To verify that the efficiency of GA-PSACO algorithm in obstacle avoidance path planning is indeed improved, three sets of simulation control experiments are set up here. The analysis of the relationship between steering angle and displacement shows that the greater the angle, the greater the influence on displacement. Therefore, the angle of the heuristic function is set to 0°, 45°, 90° and 135° The maximum number of iterations of the algorithm is set to 60 and the parameters and weights are set to 2. The generation value of each mesh point is set to 1 for all adjacent unobstructed meshes and 0 for obstructed meshes and non-adjacent meshes. The process of simulation analysis is to first input the raster map and initialize the parameters of the algorithm. The algorithm is then used to find the shortest path, and if an ant does not make it to the end, the ant's path length is denoted as 0. After the algorithm is finished, the trajectory of the shortest path and the curve of the length of the shortest path with the number of iterations are drawn according to the current state.

The obstacle grid map built for the simulation experiments is shown in Fig. 5, with horizontal and vertical coordinates from 0 to 20. Each square in the map needs to be marked with a serial number to facilitate algorithm identification and calculation. The obstacle raster map is inputted into the ant colony algorithm (ACO), the ant colony algorithm with smoothing factor introduced (Path smoothing ant colony optimization, PSACO) and the GA-PSACO algorithm to plan the shortest collision-free path from the starting point (1, 1) to the ending point (20, 20), respectively. The results are shown in Fig. 5.

In Fig. 5, the red dashed line, yellow dotted line, and blue solid line are the paths planned by GA-PSACO, PSACO, and ACO algorithms, respectively. As shown by the data, the shortest path obtained by all three algorithms is 30.38, which indicates that the GA-PSACO algorithm is effective in path planning. Comparing the paths obtained by the three algorithms with the connecting line from the starting point to the end

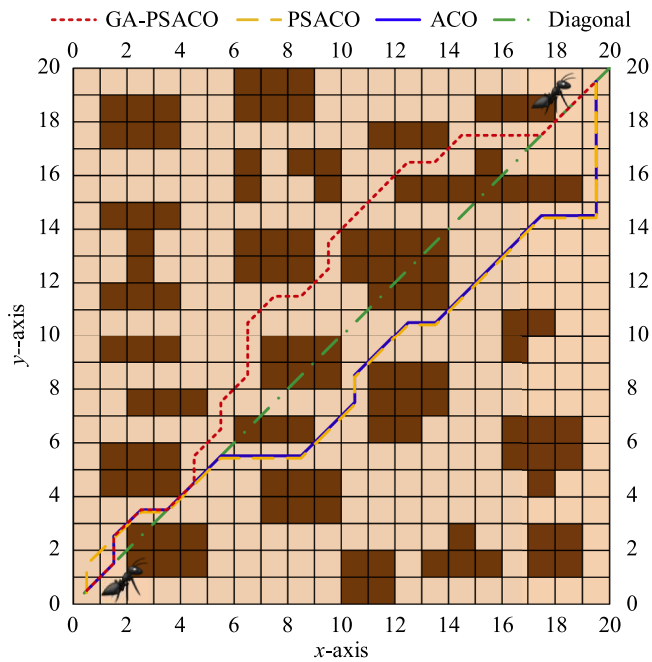


Fig. 5. Shortest collision-free paths planned under different algorithms.

point in Fig. 5, it can be seen that the paths planned by GA-PSACO are all above a straight line and that the algorithm prefers to find a path close to a straight line between the starting point and the end point. In terms of the total route, its arc is smaller and smoother. While the paths planned by ACO and PSACO algorithms are distributed on both sides of the straight line, which indicates that these two algorithms are more random and disorderly in the selection of paths during the operation, with little purpose, which will reduce the operating efficiency of the algorithms. The relationship curve between the number of iterations and the shortest path shown in Fig. 6 can illustrate the advantages of the GA-PSACO algorithm more intuitively.

As can be seen from Fig. 6, the shortest collision-free path found by the algorithm finally stabilizes at 30.38 as the number of iterations increases, and the path length does not change. However, the number of iterations used by the ACO, PSACO, and GA-PSACO algorithms to find the shortest path differs, which are 24, 17, and 4 iterations, respectively.

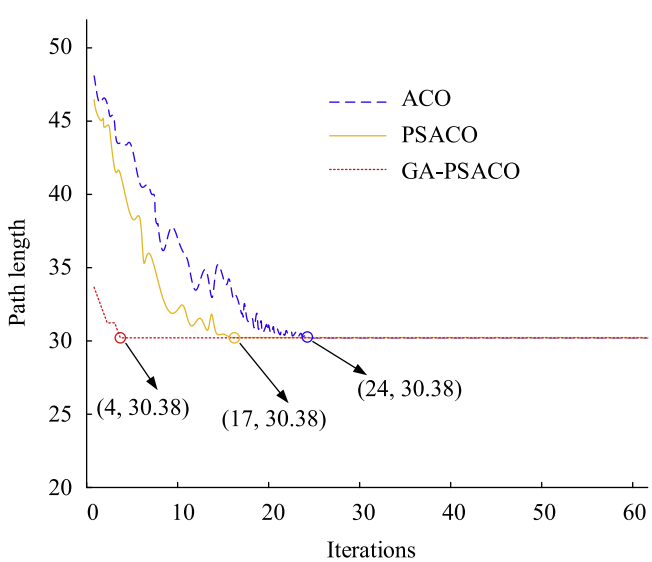


Fig. 6. Relationship curve between the number of iterations and the shortest path.

Comparing the ACO and PSACO algorithms, we can find that the PSACO algorithm with the introduction of the smoothing factor and the bootstrap factor requires fewer iterations than the ACO algorithm, and its operational efficiency is improved. The GA-PSACO algorithm, which introduces both smoothing and guidance factors and incorporates genetic algorithm, has fewer iterations, indicating that it can find the optimal path in shorter running time, which greatly improves the efficiency of the algorithm in solving the shortest collision-free path. To further verify the performance of the GA-PSACO algorithm in multi-objective point route planning, the Eil51 data set in the TSPLIB test set is used to experiment the algorithm and compare it with the ACO and PSACO algorithms. The convergence speed of the three algorithms is shown in Fig. 7.

The datas in Fig. 7 show that the shortest paths planned by the ACO, PSACO, and GA-PSACO algorithms are 442.54, 433.26, and 428.39, respectively, while the value given for the Eil51 dataset is 427. The comparison shows that the GA-PSACO algorithm obtains the smallest results among the three algorithms, which indicates that the GA-PSACO algorithm is better adapted to the Eil51 dataset. Combined with the convergence analysis in Fig. 7, the number of iterations used by the ACO, PSACO, and GA-PSACO algorithms to find the shortest path is 283, 164, and 78, respectively, and the GA-PSACO algorithm has the fastest convergence speed, indicating that the algorithm can still The GA-PSACO algorithm has the fastest convergence speed, which indicates that the algorithm can still maintain excellent performance in path planning with multiple objective points. Overall, the GA-PSACO algorithm can find the optimal paths quickly for both single-objective and multi-objective path planning. Fig. 8 visually displays the planning circuit diagrams of Eil51 using three algorithms.

From the comparison in Fig. 8, it can be seen that the path planned by the GA-PSACO algorithm is significantly closer to the standard path than the other two algorithms, further verifying the accuracy of the algorithm's calculation results. Among them, the ACO algorithm plans the longest path, but it is smoother compared to the other three algorithms, indicating that it sacrifices the length of the path during path optimization and tends to be smoother.

The calculation time and memory of each algorithm are compared, and the comparison results are shown in Table 1. The calculation time of GA-PSACO in the three data sets is 9.13 ms, 10 ms and 10.02 ms respectively, which is less than other algorithms. The memory used by this algorithm is 6.36 MB, 8.56 MB and 8.66 MB respectively, which is smaller than other algorithms. It can be seen that the algorithm in this study has the fastest running speed and the smallest memory.

Application and effect evaluation of tourism route planning model

In recent years, Guangxi tourist attractions have received much enthusiasm, and there are fewer studies on Guangxi tourist route planning. Therefore, the experiment takes Guangxi as the research object,

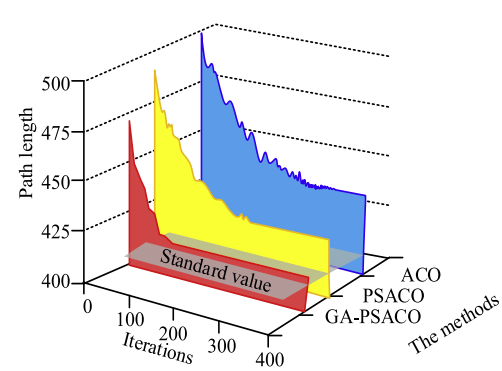


Fig. 7. Convergence of three algorithms in multi-objective point route planning.

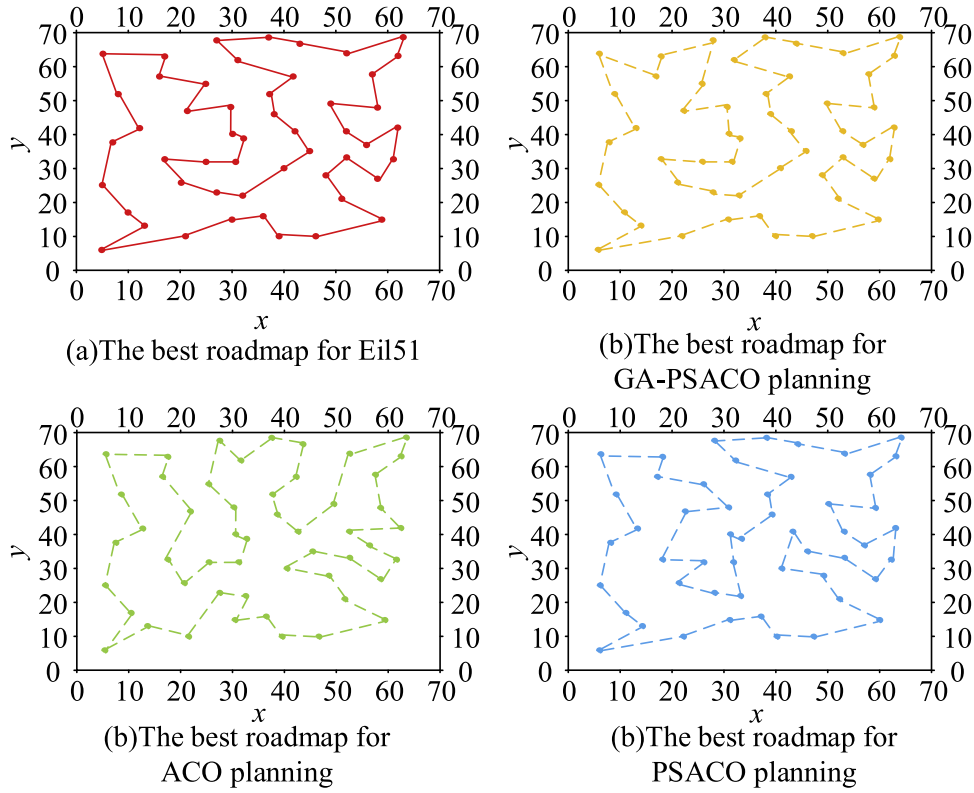


Fig. 8. Standard path and three algorithm planned paths.

Table 1
Calculation time and memory of each algorithm.

Dataset complexity	Index	ACO	PSACO	GA-PSACO
Simple	Computing time	13ms	13.05ms	9.13ms
	Memory used	11.86MB	11.63MB	6.36MB
More Complex	Computing time	14ms	15ms	10ms
	Memory used	12.85MB	12.08MB	8.56MB
Most complex	Computing time	21.24ms	13ms	10.02ms
	Memory used	13.55MB	12.30MB	8.66MB

sets each city as the tourist destination, and plans the tourist routes that tourists are satisfied by GA-PSACO algorithm. The geographic information needed for the experiment was obtained from the China Basic Geographic Information Center. Firstly, the map of each city in Guangxi is reduced in equal proportion to the range of 40×70 raster map, and the position of each city is replaced by the position of the city center point to get the horizontal and vertical coordinates of each point, as shown in Fig. 9.

According to the traffic condition of each city in Guangxi, Vij is assigned to the travel speed between cities respectively. According to the survey of the number of tourists received in Guangxi, the number of tourists in Beihai, Nanning, Guilin and Liuzhou is more, so the number of tourists is assigned to 2 million for the convenience of calculation, while the number of tourists in other cities is assigned to 1 million if they are less. The maximum tourist carrying capacity of each place is based on the area of the city, and the maximum carrying capacity of Nanning, Guilin, Baise and Hechi with larger area can be assigned to 5 million people, while the maximum carrying capacity of the rest of the cities is assigned to 4 million people. The tourism scores of each city in Guangxi and the similarity of some tourists' tourism preference characteristics with the characteristics of each city in Guangxi are obtained from the Internet, and the relevant values are shown in Table 2. The scores in the table are scored as 10 out of 10. Nanning has the highest Feature Similarity at 0.90, as well as the highest Regional score and Preference level

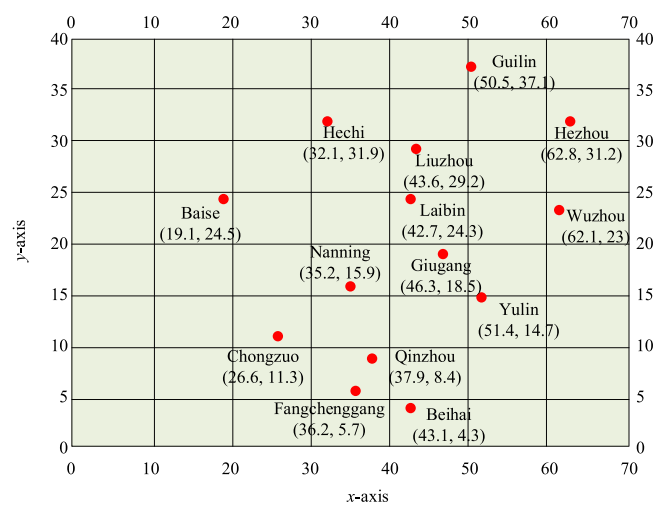


Fig. 9. Coordinate distribution map of each city in Guangxi.

scores. It shows that tourists' preference for Nanning is the highest among all cities in Guangxi, and the priority of choosing Nanning as a tourist destination is the highest.

The weights ofε,τ andδ in the algorithm are set to 2, and the tour time is set to 30 days. According to the GA-PSACO algorithm, the tourist route map in Guangxi can be output, as shown in Fig. 10.

Fig. 10(a) shows the optimal travel path and the best visit time in Guangxi planned according to some tourists' preference features. The total length of the path is calculated to be 142.74. According to the relationship between the number of iterations and the path length variation shown in Fig. 10(b), it can be seen that the GA-PSACO algorithm completes the optimal path planning and stabilizes in only three iterations. It indicates that the efficiency of the algorithm in travel path

Table 2
Preference of tourists for cities in Guangxi.

City	Feature similarity	Regional score	Preference level
Nanning	0.90	8.23	7.40
Liuzhou	0.85	8.15	6.92
Guilin	0.88	8.13	7.16
Wuzhou	0.54	8.10	4.39
Beihai	0.89	8.01	7.13
Fangchenggang	0.58	8.14	4.75
Qinzhou	0.53	7.84	4.18
Yulin	0.57	7.94	4.54
Baise	0.67	8.19	5.46
Hezhou	0.55	7.99	4.41
Hechi	0.43	7.90	3.42
Chongzuo	0.45	7.67	3.45
Laibin	0.40	8.07	3.23
Guigang	0.50	7.57	3.79

planning can still maintain a high level even with the addition of constraints. To verify the superiority of the GA-PSACO algorithm, the results shown in Fig. 11 can be obtained by comparing it with the traditional genetic algorithm (GA) and simulated annealing algorithm (SAA), using tourist satisfaction as the evaluation index.

As shown in Fig. 11, the average satisfaction rate of the proposed model is 15.38, and the average satisfaction rates of the GA and SAA algorithms are 13.54 and 11.76, respectively. The tourists' satisfaction rate with the routes planned by the GA-PSACO algorithm tends to be stable, especially in the late stage of the trip, and remains high. On the other hand, the tourists' satisfaction with the routes planned by the other two algorithms is lower and fluctuates more than that of the GA-PSACO algorithm, both in the early and in the late stages of the trip. The analysis may be due to the fact that after the introduction of genetic algorithm to optimize the pheromone initialization, the model has more diverse individuals, and combined with the improved transfer probability calculation formula, it can improve the adaptability of the model for path finding and enhance the global finding ability, so that the GA-PSACO algorithm can consider the tourists' journey itinerary more comprehensively. In a comprehensive view, the GA-PSACO algorithm performs better in improving the tour experience and can take into account tourists' feelings from multiple perspectives, such as congestion avoidance, tourists' preferences and travel time, so that they can obtain a better travel experience. In order to explore the performance advantages of this algorithm in tourism path planning, it is compared with three Ant colony optimization algorithms researched by Mutar et al. (2020), Shao et al. (2020), Zheng et al. (2020), and the artificial potential field and fuzzy logic algorithm that are widely used in current path planning. The results of minimum path length, customer satisfaction, running time, and iteration times are shown in Table 3.

From Table 3, it can be seen that among the above algorithms, GA-PSACO performs the best in all evaluation indicators. Compared with the other four combination algorithms, the single algorithm of artificial potential field and fuzzy logic has the worst application effect in tourism route planning, with the minimum path length being above 150, indicating that the combination algorithm has more advantages in multi-objective path planning. Compared with the Ant colony optimization

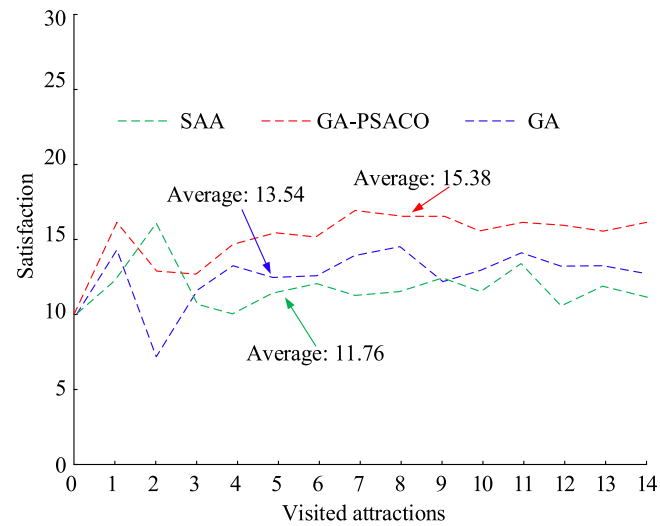
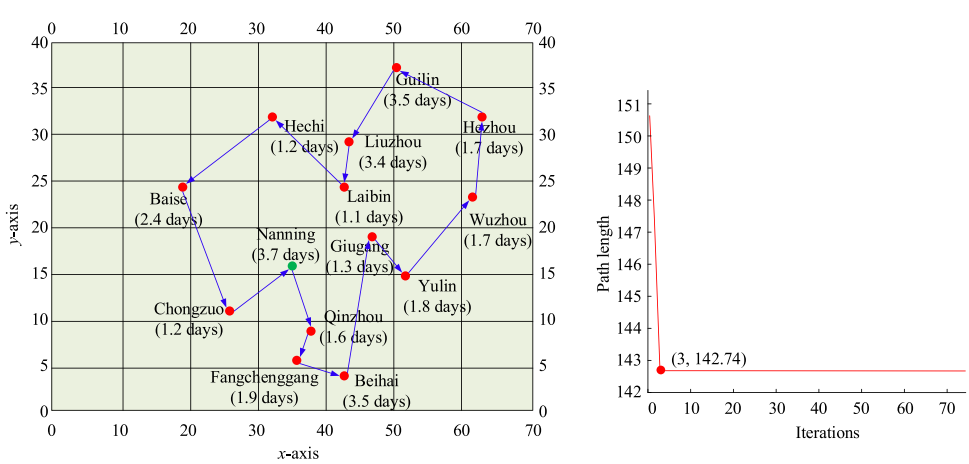


Fig. 11. Visitor satisfaction curve.

Table 3
Results of different algorithms for tourism route planning.

Algorithm type	Minimum path length	Customer satisfaction	Running time(s)	Iterations
Mutar et al. (Mutar et al., 2020)	147.13	11.16	51	29
Shao et al. (Shao et al., 2020)	145.42	12.43	23	14
Zheng et al. (Zheng et al., 2020)	144.71	13.27	43	21
Artifificial potential fifield	153.36	9.54	72	49
Fuzzy logic	159.25	9.20	67	43
GA-PSACO	142.74	15.38	6	3



(a) The best tourist route and time in Guangxi (b) The path change curve

Fig. 10. GA-PSACO algorithm for tourism route planning in Guangxi.

algorithms studied by Mutar et al. (2020), Shao et al. (2020), Zheng et al. (2020), GA-PSACO not only has the optimal operation result, but also has greatly improved the operation efficiency, which indicates that the model has better performance in tourism route planning.

Conclusions and future works

As internet technology matures and tourism develops, the way tourists travel has also changed. Smart tourism is popular because it can recommend travel routes according to tourists' preferences. Tourist route planning has also become a hot topic in tourism research. To address this problem, this paper proposes an intelligent travel route planning model based on efficiency improvement.Based on ant colony algorithm, the model integrates the individual update and cross operation of genetic algorithm, and changes the calculation method of transfer probability, and obtains GA-PSACO algorithm. The algorithm is simulated and analyzed, and then the model is applied to Guangxi tourism route planning. The simulation results show that GA-PSACO algorithm can find the shortest path in the simple obstacle avoidance route planning, and only iterates 4 times to complete the path finding. The computation time of GA-PSACO in three data sets is 9.13 ms, 10 ms and 10.02 ms respectively, which is less than other algorithms. The memory used by this algorithm is 6.36 Mb, 8.56 Mb and 8.66 Mb respectively, which is smaller than the other algorithms. It can be seen that this algorithm has the fastest running speed and smaller memory. It shows that the GA-PSACO algorithm can find the optimal solution quickly, whether it is single objective point path planning or multi-objective point path planning. In the practical application evaluation, the model can plan the travel route quickly, and the average satisfaction is 15.38, which is higher than other models. It shows that the GA-PSACO algorithm can consider the whole trip of tourists more comprehensively, so that tourists can have a better tour experience. The overall performance of the model is good, and the convergence rate is faster. In this study, ant colony algorithm is applied to tourist route planning, which makes full use of many advantages of ant colony algorithm, but how to effectively use ant colony algorithm to solve other problems is worth further study. In addition, the dynamic environment or other practical conditions will also have various influences on the application of ant colony algorithm, but these conditions are not analyzed in this study. Future work will attempt to expand the application of the ant colony algorithm by applying it to different industries, such as logistics and transportation, and will take into account how the environment may affect the algorithm.

Fundings

The research is supported by: Key scientific research projects of colleges and universities in Henan province, Structural Model Building and strategy analysis of Henan digital tourism industryrom the perspective of high-quality development, (No.23A630038).

Credit author statement

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Shuping Liu, Hao Wu. The first draft of the manuscript was written by Shuping Liu and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors have no relevant financial or non-financial interests to disclose.

Data availability

No data was used for the research described in the article.

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